

Equilibrium Operators, Variational Inequalities and Welfare:

What Is and Is Not a Theorem

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Abstract

Research line EQO001 asked whether a formal relationship between equilibrium operators, variational inequalities and welfare criteria could yield results beyond the existing ones; its public record calls the formulation reviewed a combination of known components without an independent theorem. This note records that assessment. With equilibria the solutions of $VI(F, K)$ on a closed convex set K and welfare a C^1 functional on K , we prove the four classical facts on which any such relationship rests: potentials, monotonicity, the equilibrium–welfare gap (Pigou, 4/3) and the first-order coincidence whose repair is marginal-cost pricing. The one small addition characterises the cone $\Lambda(x^*)$ of welfare weights for which a Nash equilibrium maximises $\sum_i \lambda_i u_i$: a closed convex cone inside a first-order cone $\Lambda_1(x^*)$ built from $F(x^*)$, the normal cone and the marginal externalities, with equality under joint concavity; (F, K) alone never determines it. Two examples show that $\Lambda(x^*)$ need not be polyhedral and that $\Lambda \neq \Lambda_1$ in two-commodity routing with affine costs. Numerical checks on 400 random monotone games confirm equality in every tested direction of every concave instance and certify strict inclusion in at least 56% of the non-concave instances with a non-trivial cone. No new equilibrium concept is offered.

1 Introduction

The variational-inequality formulation of equilibrium is one of the unifying ideas of applied mathematics: a Wardrop traffic equilibrium [29], a Nash equilibrium of a concave game [21, 25] and a Walrasian or spatial price equilibrium can all be written as: find x^* in a closed convex set K with $\langle F(x^*), x - x^* \rangle \geq 0$ for all $x \in K$ [6, 10, 12, 14, 16, 20]. Welfare is an optimisation problem over the same K , and the temptation that motivated research line EQO001 is obvious: to look for a “formal relationship” between the inequality and the optimisation problem from which welfare statements about equilibria would follow.

The public record of the line is candid: the formulation reviewed combined known components without establishing an independent theorem, and the line was paused. This note discharges the line’s objective in the only way that seems honest to us: by writing down, with complete proofs where they are short, what is a theorem in this area, what is a definition in disguise, and what remains open. We add one small result on the set of welfare weights that make a given equilibrium optimal; Remark 5.2 states which parts of it are standard and which are, as far as we could check, only repackaged here.

Contributions and scope. Section 2 fixes the setting; Section 3 proves the classical results; Section 4 gives the first-order characterisation of welfare-optimal equilibria and marginal-cost pricing; Section 5 contains the welfare-weight cone (Theorem 5.1), the impossibility statement

(Proposition 5.3) and four exact examples, two of them negative (Examples 5.6, 5.7); Sections 6–8 report the numerical checks, the status of every claim, limitations and two open questions. Everything is finite-dimensional and deterministic; every number in the text is produced by the scripts of Appendix A with fixed seeds.

2 Setting

Throughout, $K \subseteq \mathbb{R}^n$ is a nonempty closed convex set, $U \supseteq K$ an open set and $F: U \rightarrow \mathbb{R}^n$ a continuous map, the *equilibrium operator*. We write $\langle \cdot, \cdot \rangle$ for the Euclidean inner product.

Definition 2.1 (Variational inequality, normal cone). $\text{VI}(F, K)$ is the problem of finding $x^* \in K$ such that $\langle F(x^*), x - x^* \rangle \geq 0$ for all $x \in K$; its solution set is $\text{SOL}(F, K)$. The *normal cone* of K at $x \in K$ is $N_K(x) = \{v \in \mathbb{R}^n : \langle v, y - x \rangle \leq 0 \ \forall y \in K\}$, a closed convex cone, equal to $\{0\}$ when x is an interior point. By definition,

$$x^* \in \text{SOL}(F, K) \iff -F(x^*) \in N_K(x^*). \quad (1)$$

Definition 2.2 (Monotonicity). F is *monotone* on K if $\langle F(x) - F(y), x - y \rangle \geq 0$ for all $x, y \in K$; *strictly monotone* if the inequality is strict for $x \neq y$; μ -*strongly monotone* if $\langle F(x) - F(y), x - y \rangle \geq \mu \|x - y\|^2$ with $\mu > 0$; and L -*Lipschitz* if $\|F(x) - F(y)\| \leq L \|x - y\|$.

Definition 2.3 (Game). A *game* has N players, strategy sets $K_i \subseteq \mathbb{R}^{n_i}$ closed and convex, $K = K_1 \times \dots \times K_N$, and payoffs $u_i: U \rightarrow \mathbb{R}$ of class C^1 on an open $U \supseteq K$, with $u_i(\cdot, x_{-i})$ concave on K_i for every x_{-i} . A *Nash equilibrium* is $x^* \in K$ with $u_i(x^*) \geq u_i(y_i, x_{-i}^*)$ for all i and $y_i \in K_i$. The *pseudo-gradient* (equilibrium operator) is $F(x) = (-\nabla_{x_1} u_1(x), \dots, -\nabla_{x_N} u_N(x))$, where ∇_{x_i} is the gradient with respect to player i 's own variables. The *marginal externality* of player j on player $i \neq j$ at x is $\nabla_{x_j} u_i(x) \in \mathbb{R}^{n_j}$.

Lemma 2.4 (Nash equilibria are VI solutions). *In the setting of Definition 2.3, x^* is a Nash equilibrium if and only if $x^* \in \text{SOL}(F, K)$. [classical, 25; 10]*

Proof. For a concave differentiable function ϕ on a convex set C , z maximises ϕ over C iff $\langle \nabla \phi(z), y - z \rangle \leq 0$ for all $y \in C$: necessity because $t \mapsto \phi(z + t(y - z))$ is defined on $[0, 1]$ and has right derivative $\langle \nabla \phi(z), y - z \rangle$ at 0, which must be ≤ 0 ; sufficiency because $\phi(y) \leq \phi(z) + \langle \nabla \phi(z), y - z \rangle$. Apply this to $\phi = u_i(\cdot, x_{-i}^*)$ on K_i : x^* is Nash iff $\langle -F_i(x^*), y_i - x_i^* \rangle \leq 0$ for all i and $y_i \in K_i$. Summing over i gives $\langle F(x^*), y - x^* \rangle \geq 0$ for every $y \in K$; conversely, choosing $y = (y_i, x_{-i}^*)$ in the VI recovers the i -th inequality. $\square$

Definition 2.5 (Nonatomic routing). A network has edges E , commodities $k = 1, \dots, m$ with demands $d_k > 0$ and path sets P_k , $P = \bigsqcup_k P_k$. A path flow $f \in \mathbb{R}_+^P$ is feasible if $\sum_{p \in P_k} f_p = d_k$ for each k ; K is the set of feasible flows, a product of scaled simplices. Edge flows are $f_e = \sum_{p \ni e} f_p$, edge costs $c_e: \mathbb{R}_+ \rightarrow \mathbb{R}_+$ are continuous and nondecreasing, path costs are $c_p(f) = \sum_{e \in p} c_e(f_e)$, and the equilibrium operator is $F(f) = (c_p(f))_{p \in P}$. A *Wardrop equilibrium* is a feasible f^* such that, for every k and $p, q \in P_k$ with $f_p^* > 0$, $c_p(f^*) \leq c_q(f^*)$. The *social cost* is $C(f) = \sum_p f_p c_p(f) = \sum_e f_e c_e(f_e)$, and welfare is $W = -C$.

Lemma 2.6 (Normal cone of a face of a simplex). *Let $\Delta = \{x \in \mathbb{R}_+^P : \sum_{p \in P_k} x_p = d_k, \ k = 1, \dots, m\}$ and $x \in \Delta$ with support $S = \{p : x_p > 0\}$. Then*

$$N_\Delta(x) = \{v : \exists \gamma \in \mathbb{R}^m \text{ with } v_p = \gamma_k \text{ for } p \in S \cap P_k, \ v_p \leq \gamma_k \text{ for } p \in P_k \setminus S\}.$$

Proof. For $p, q \in P_k$ with $q \in S$ and $t > 0$ small, $y = x + t(e_p - e_q) \in \Delta$, so $v \in N_\Delta(x)$ implies $v_p \leq v_q$; exchanging p, q when both are in S gives equality on $S \cap P_k$, say $v_p = \gamma_k$, and $v_p \leq \gamma_k$ off S . Conversely, if v has this form and $y \in \Delta$, then $\langle v, y - x \rangle = \sum_k (\sum_{p \in P_k} v_p y_p - \gamma_k d_k) \leq \sum_k (\gamma_k \sum_{p \in P_k} y_p - \gamma_k d_k) = 0$, using $v_p \leq \gamma_k$ on P_k and $y \geq 0$. $\square$

Lemma 2.7 (Wardrop equilibria are VI solutions). f^* is a Wardrop equilibrium iff $f^* \in \text{SOL}(F, K)$ with $F = (c_p)_p$. [classical, 27; 6]

Proof. By (1) and Lemma 2.6, $f^* \in \text{SOL}(F, K)$ iff for each k there is γ_k with $c_p(f^*) = -\gamma_k$ on used paths of P_k and $c_p(f^*) \geq -\gamma_k$ on unused ones, which is Wardrop's condition. $\square$

Definition 2.8 (Welfare functionals). A *welfare functional* is a C^1 function $W: U \rightarrow \mathbb{R}$ to be maximised over K . For a game, the *weighted welfare* with weights $\lambda \in \mathbb{R}_+^N$ is $W_\lambda = \sum_i \lambda_i u_i$; $\lambda = \mathbf{1}$ is the utilitarian criterion. For routing, $W = -C$.

3 The classical results, with proofs

3.1 Potentials

Proposition 3.1 (Poincaré lemma on a convex set). Let U be open and convex and $F: U \rightarrow \mathbb{R}^n$ of class C^1 with symmetric Jacobian $DF(x)$ at every x . Fix $x_0 \in U$ and set $P(x) = \int_0^1 \langle F(x_0 + t(x - x_0)), x - x_0 \rangle dt$. Then $P \in C^2(U)$ and $\nabla P = F$. [classical; proved here]

Proof. With $\gamma_t = x_0 + t(x - x_0)$ and $h = x - x_0$, differentiation under the integral sign and the symmetry $\partial_i F_j = \partial_j F_i$ give $\partial_i P(x) = \int_0^1 [F_i(\gamma_t) + t \sum_j \partial_j F_i(\gamma_t) h_j] dt = \int_0^1 \frac{d}{dt} [t F_i(\gamma_t)] dt = F_i(x)$. $\square$

Theorem 3.2 (Equilibria of gradient operators). Suppose $F = \nabla P$ on $U \supseteq K$ with $P \in C^1(U)$. Then

$$\text{SOL}(F, K) = \{x \in K : \langle \nabla P(x), y - x \rangle \geq 0 \ \forall y \in K\},$$

the set of stationary points of P over K ; it contains $\arg \min_K P$, and equals it when P is convex on K . If P is strictly convex, $\text{SOL}(F, K)$ has at most one point. [classical: 2 for routing, 18 for potential games; proved here]

Proof. The first identity is Definition 2.1 with $F = \nabla P$. If x minimises P over K then, as in the proof of Lemma 2.4, the right derivative of $t \mapsto P(x + t(y - x))$ at 0 is nonnegative, which is the stationarity inequality. If P is convex and x is stationary, $P(y) \geq P(x) + \langle \nabla P(x), y - x \rangle \geq P(x)$ for all $y \in K$. If P is strictly convex its minimiser over the convex set K is unique. $\square$

Remark 3.3 (The three classical instances). (a) *Routing* [2]. For continuous c_e , $P(f) = \sum_e \int_0^{f_e} c_e(s) ds$ is C^1 with $\partial P / \partial f_p = \sum_{e \in p} c_e(f_e) = c_p(f)$ by the fundamental theorem of calculus (no differentiability of c_e is needed), and convex when the c_e are nondecreasing; Wardrop equilibria are the minimisers of P , and the edge flows are unique when all c_e are strictly increasing. Note that $P \neq C$: the potential integrates the cost, the social cost multiplies it; this is the whole of Section 3.3. (b) *Concave games* [25]. Lemma 2.4 and Theorem 3.5(a) give uniqueness under Rosen's diagonal strict concavity, which for unit weights is strict monotonicity of the pseudo-gradient. (c) *Potential games* [18]. A C^1 game is an exact potential game with potential Π iff $F = -\nabla \Pi$; its equilibria are the stationary points of $-\Pi$ over K , its maximisers when Π is concave. The potential is bookkeeping, not welfare: in general $\Pi \neq \sum_i u_i$ (Example 5.5(i) is a potential game whose equilibrium maximises no weighted welfare).

3.2 Monotonicity: uniqueness, existence and a solver

Lemma 3.4 (Projection). Let Π_K be the Euclidean projection onto K . Then $z = \Pi_K(y)$ iff $z \in K$ and $\langle y - z, x - z \rangle \leq 0$ for all $x \in K$; Π_K is nonexpansive; and for every $\gamma > 0$, $x \in \text{SOL}(F, K)$ iff $x = \Pi_K(x - \gamma F(x))$. [classical, 16, Chapter I]

Proof. The first claim is Theorem 3.2 for the strictly convex $P = \frac{1}{2} \|y - \cdot\|^2$; adding the characterisations at $z = \Pi_K y$ and $z' = \Pi_K y'$ gives $\|z - z'\|^2 \leq \langle y - y', z - z' \rangle$, hence nonexpansiveness; and $x = \Pi_K(x - \gamma F(x))$ iff $\langle -\gamma F(x), y - x \rangle \leq 0$ for all $y \in K$, which is the VI. $\square$

Theorem 3.5 (Monotone operators). (a) If F is strictly monotone on K , $\text{SOL}(F, K)$ has at most one point. (b) If F is μ -strongly monotone and L -Lipschitz on K , $\text{SOL}(F, K)$ is a singleton $\{x^*\}$, and for $0 < \gamma < 2\mu/L^2$ the iteration $x_{t+1} = \Pi_K(x_t - \gamma F(x_t))$ satisfies $\|x_{t+1} - x^*\| \leq q\|x_t - x^*\|$ with $q = \sqrt{1 - 2\gamma\mu + \gamma^2 L^2} < 1$; for $\gamma = \mu/L^2$, $q = \sqrt{1 - \mu^2/L^2}$. [classical, 10, 16; proved here]

Proof. (a) If $x, y \in \text{SOL}(F, K)$ then $\langle F(x), y - x \rangle \geq 0$ and $\langle F(y), x - y \rangle \geq 0$; adding, $\langle F(x) - F(y), x - y \rangle \leq 0$, so $x = y$ by strict monotonicity. (b) Let $T(x) = \Pi_K(x - \gamma F(x))$. By nonexpansiveness, $\|Tx - Ty\|^2 \leq \|(x - y) - \gamma(F(x) - F(y))\|^2 = \|x - y\|^2 - 2\gamma\langle F(x) - F(y), x - y \rangle + \gamma^2\|F(x) - F(y)\|^2 \leq q^2\|x - y\|^2$, and $q < 1$ iff $\gamma < 2\mu/L^2$. Banach's fixed-point theorem on the complete set K gives a unique fixed point, which is the unique VI solution by Lemma 3.4, and the contraction estimate is the stated rate. $\square$

Remark 3.6. For merely continuous F on compact convex K , existence is the Hartman–Stampacchia theorem (14, Lemma 3.1), obtained from Brouwer's theorem applied to T ; we do not reproduce it. All solvers in Section 6 are the iteration of Theorem 3.5(b) with $\gamma = \mu/L^2$, so each computed equilibrium comes with a uniqueness proof and a residual certificate.

3.3 Equilibrium is not welfare: Pigou and the price of anarchy

Example 3.7 (Pigou). Two parallel links, unit demand, $c_1 \equiv 1$ and $c_2(x) = x^d$, $d \geq 1$ [23]. Every user on link 2 pays $1 \leq c_1$, so $f^* = (0, 1)$ is a Wardrop equilibrium with $C(f^*) = 1$, and it is the only one: a feasible flow with $f_1 > 0$ has $f_2 < 1$, so its users on link 1 pay $1 > f_2^d$. The social cost of $(1 - x, x)$ is $1 - x + x^{d+1}$, minimised at $x = (d+1)^{-1/d}$ with value $1 - \frac{d}{d+1}(d+1)^{-1/d}$. For $d = 1$ the optimum is $(\frac{1}{2}, \frac{1}{2})$ with cost $\frac{3}{4}$ and ratio $\frac{4}{3}$; the ratio tends to infinity with d , since $(d+1)^{-1/d} \rightarrow 1$. The computed values are in Figure 2 and `results/tables_traffic.md`.

Theorem 3.8 (Price of anarchy for affine costs). If every $c_e(x) = a_e x + b_e$ with $a_e, b_e \geq 0$, then for every Wardrop equilibrium f and every feasible f' , $C(f) \leq \frac{4}{3}C(f')$. The constant is attained in Example 3.7 with $d = 1$. [classical, 26; the proof below follows 5]

Proof. By Lemma 2.7, $\sum_p c_p(f)(f'_p - f_p) \geq 0$; regrouping by edges, $C(f) = \sum_e c_e(f_e)f_e \leq \sum_e c_e(f_e)f'_e$. For each edge, $c_e(f_e)f'_e = c_e(f'_e)f'_e + a_e(f_e - f'_e)f'_e \leq c_e(f'_e)f'_e + \frac{1}{4}a_e f_e^2 \leq c_e(f'_e)f'_e + \frac{1}{4}c_e(f_e)f_e$, since $y \mapsto a_e(f_e - y)y$ is maximised at $y = f_e/2$ and $a_e f_e^2 \leq (a_e f_e + b_e)f_e$. Summing, $C(f) \leq C(f') + \frac{1}{4}C(f)$. $\square$

4 When an equilibrium is a welfare maximiser

Proposition 4.1 (Necessary and sufficient first-order conditions). Let $x^* \in \text{SOL}(F, K)$ and let W be differentiable at x^* .

- (a) If $x^* \in \arg \max_K W$ then $\nabla W(x^*) \in N_K(x^*)$ and $-F(x^*) \in N_K(x^*)$. If W is concave on K , the converse holds: $x^* \in \arg \max_K W$ iff $\nabla W(x^*) \in N_K(x^*)$.
- (b) If x^* is an interior point of K , (a) forces $F(x^*) = 0 = \nabla W(x^*)$. If $N_K(x^*) = \mathbb{R}_+\nu$ is a ray and $F(x^*) \neq 0$, (a) forces $\nabla W(x^*) = -\theta F(x^*)$ for some $\theta \geq 0$.
- (c) (Standard form.) Let $K = \{x \in \mathbb{R}_+^n : Ax = b\}$ with $A \in \mathbb{R}^{m \times n}$ and let S be the support of x^* . If x^* is an equilibrium and a W -maximiser, there are $y, z \in \mathbb{R}^m$ with

$$\begin{aligned} F_S(x^*) &= (A^\top y)_S, & F_j(x^*) &\geq (A^\top y)_j \quad (j \notin S), \\ \nabla_S W(x^*) &= (A^\top z)_S, & \partial_j W(x^*) &\leq (A^\top z)_j \quad (j \notin S). \end{aligned}$$

In particular, on a single simplex ($A = \mathbf{1}^\top$) both $F(x^*)$ and $\nabla W(x^*)$ are constant on the support (hence parallel on it when both are non-zero), and each satisfies Wardrop's principle with respect to its own coordinates.

(d) If F is strictly monotone, W is strictly concave and both $\text{SOL}(F, K)$ and $\arg \max_K W$ are nonempty, then $\text{SOL}(F, K) = \arg \max_K W$ iff the unique equilibrium satisfies $\nabla W(x^*) \in N_K(x^*)$; otherwise the two sets are disjoint.

[proved here; elementary]

Proof. (a) The first-order necessary condition for a maximum of a differentiable function over a convex set (proof of Lemma 2.4) is $\langle \nabla W(x^*), y - x^* \rangle \leq 0$ for all $y \in K$, i.e. $\nabla W(x^*) \in N_K(x^*)$; the second membership is (1); sufficiency under concavity is the same argument as in Theorem 3.2. (b) For interior x^* , $N_K(x^*) = \{0\}$. If $N_K(x^*) = \mathbb{R}_+ \nu$, write $-F(x^*) = \alpha \nu$ and $\nabla W(x^*) = \beta \nu$ with $\alpha, \beta \geq 0$; $F(x^*) \neq 0$ gives $\alpha > 0$ and $\theta = \beta/\alpha$. (c) The tangent cone of K at x^* is $\{h : Ah = 0, h_j \geq 0 \text{ (} j \notin S \text{)}\}$ and its polar, by Farkas' lemma [24], is $N_K(x^*) = \{A^\top w - v : w \in \mathbb{R}^m, v \geq 0, v_S = 0\}$. Apply this to $-F(x^*)$ (with $y = -w$) and to $\nabla W(x^*)$ (with $z = w$). (d) Both sets are singletons by Theorem 3.5(a) and strict concavity; apply (a). $\square$

Proposition 4.1 is a statement about first-order conditions and nothing more: it says that an equilibrium can be a welfare optimum only if the two problems happen to share their KKT point, and it gives no mechanism producing that coincidence. The one mechanism that always works is to change the operator.

Corollary 4.2 (Marginal-cost pricing and the toll set). *Let $C \in C^1(U)$ be a cost to be minimised over K and define the toll field $\tau = \nabla C - F$. Then*

$$\text{SOL}(F + \tau, K) = \{x \in K : -\nabla C(x) \in N_K(x)\} \supseteq \arg \min_K C,$$

with equality when C is convex on K , and at most one point when C is strictly convex; the tolled operator is a gradient field with potential C . For separable routing costs with $c_e \in C^1$, $\tau_p(f) = \sum_{e \in P} f_e c'_e(f_e)$, the Pigou–Knight–Beckmann marginal-cost toll. More generally, for a target $\bar{x} \in K$ the set of constant toll vectors $t \in \mathbb{R}^n$ with $\bar{x} \in \text{SOL}(F + t, K)$ is the translated cone $-F(\bar{x}) - N_K(\bar{x})$, polyhedral when K is; choosing in it the toll closest to a reference vector is an inverse optimisation problem in the sense of Ahuja and Orlin [1], and for routing it is the toll-set description of Hearn and Ramana [15]. [classical, 2, 15, 17, 23; proved here]

Proof. $F + \tau = \nabla C$, so the first statement is Theorem 3.2 with $P = C$. For routing, $\partial C / \partial f_p = \sum_{e \in P} (c_e(f_e) + f_e c'_e(f_e))$, whence τ_p . Finally $\bar{x} \in \text{SOL}(F + t, K)$ iff $-F(\bar{x}) - t \in N_K(\bar{x})$ by (1). $\square$

5 The cone of welfare weights

Fix a game as in Definition 2.3 and a Nash equilibrium x^* . Write $u = (u_1, \dots, u_N) : K \rightarrow \mathbb{R}^N$ and $G(x^*) = Du(x^*) \in \mathbb{R}^{N \times n}$ for the Jacobian of the payoff vector, whose i -th row is $\nabla u_i(x^*)^\top$ (all variables, own and others'). Define

$$\Lambda(x^*) = \{\lambda \in \mathbb{R}_+^N : x^* \in \arg \max_K W_\lambda\}, \quad \Lambda_1(x^*) = \{\lambda \in \mathbb{R}_+^N : G(x^*)^\top \lambda \in N_K(x^*)\}.$$

For a set $D \subseteq \mathbb{R}^N$, $D^\circ = \{\lambda : \langle \lambda, d \rangle \leq 0 \ \forall d \in D\}$ is its polar cone.

Theorem 5.1 (Welfare-weight cone). (a) $\Lambda(x^*) = \mathbb{R}_+^N \cap \{u(x) - u(x^*) : x \in K\}^\circ$. In particular $\Lambda(x^*)$ is a closed convex cone.

(b) $\Lambda(x^*) \subseteq \Lambda_1(x^*)$, and $\Lambda_1(x^*)$ is a closed convex cone, polyhedral whenever K is polyhedral.

(c) If every u_i is concave on K (jointly in all variables), then $\Lambda(x^*) = \Lambda_1(x^*)$. More generally, equality holds whenever W_λ is concave on K for every $\lambda \in \Lambda_1(x^*)$.

(d) (Block form.) $\lambda \in \Lambda_1(x^*)$ iff for every player j

$$-\lambda_j F_j(x^*) + \sum_{i \neq j} \lambda_i \nabla_{x_j} u_i(x^*) \in N_{K_j}(x_j^*). \quad (2)$$

Since $-F_j(x^*) \in N_{K_j}(x_j^*)$ already, a sufficient condition is that the λ -weighted externality $\sum_{i \neq j} \lambda_i \nabla_{x_j} u_i(x^*)$ lies in $N_{K_j}(x_j^*)$ for every j ; when x_j^* is interior to K_j , (2) says exactly that this weighted externality vanishes.

(e) $\Lambda(x^*) \neq \{0\}$ implies that x^* is weakly Pareto efficient in K (no $x \in K$ with $u_i(x) > u_i(x^*)$ for all i); under the concavity of (c), the converse holds.

[proved here; (a) and (e) are the standard scalarisation argument of multi-objective optimisation [9, 11], and $\Lambda(x^*)$ is the cell at $u(x^*)$ of the weight-set decomposition studied for linear problems by Benson and Sun [3]]

Proof. (a) $\lambda \in \Lambda(x^*)$ iff $\lambda \geq 0$ and $\langle \lambda, u(x) - u(x^*) \rangle \leq 0$ for all $x \in K$, which is the stated intersection; polar cones and the orthant are closed convex cones. (b) If x^* maximises the differentiable W_λ over K , the first-order condition gives $\nabla W_\lambda(x^*) = \sum_i \lambda_i \nabla u_i(x^*) = G(x^*)^\top \lambda \in N_K(x^*)$. $\Lambda_1(x^*)$ is the preimage of the closed convex cone $N_K(x^*)$ under a linear map, intersected with $\mathbb{R}_+^N$; when K is polyhedral so is $N_K(x^*)$, hence so is $\Lambda_1(x^*)$. (c) For $\lambda \geq 0$, W_λ is concave when the u_i are, and the first-order condition is then sufficient (Proposition 4.1(a)); the general statement is the same argument. (d) For a product set, $v \in N_K(x^*)$ iff $v_j \in N_{K_j}(x_j^*)$ for each j : taking $y \in K$ that differs from x^* in block j only shows necessity, and summing the block inequalities shows sufficiency. Block j of $G(x^*)^\top \lambda$ is $\sum_i \lambda_i \nabla_{x_j} u_i(x^*) = \lambda_j \nabla_{x_j} u_j(x^*) + \sum_{i \neq j} \lambda_i \nabla_{x_j} u_i(x^*)$ and $\nabla_{x_j} u_j(x^*) = -F_j(x^*)$. The sufficient condition follows because $N_{K_j}(x_j^*)$ is a convex cone containing $-F_j(x^*)$ by Lemma 2.4; for interior x_j^* , $N_{K_j}(x_j^*) = \{0\}$ forces $F_j(x^*) = 0$ and then (2) reads $\sum_{i \neq j} \lambda_i \nabla_{x_j} u_i(x^*) = 0$. (e) If $\lambda \in \Lambda(x^*) \setminus \{0\}$ and $u(x) > u(x^*)$ componentwise, then $\langle \lambda, u(x) \rangle > \langle \lambda, u(x^*) \rangle$, a contradiction. Conversely, under concavity the set $A = u(K) - \mathbb{R}_+^N$ is convex; weak efficiency means $u(x^*) + \varepsilon \mathbf{1} \notin A$ for every $\varepsilon > 0$, so $u(x^*)$ is a boundary point of A and a supporting hyperplane gives $\lambda \neq 0$ with $\langle \lambda, a \rangle \leq \langle \lambda, u(x^*) \rangle$ for all $a \in A$; since $u(x^*) - \mathbb{R}_+^N \subseteq A$, $\lambda \geq 0$, and $a = u(x)$ gives $\lambda \in \Lambda(x^*)$. $\square$

Remark 5.2 (What is standard and what is not). The scalarisation in (a) and (e) is textbook, and the set of weights under which a given point maximises the weighted sum is a standard object of multi-objective optimisation (the weight-set decomposition, polyhedral in the linear case [3]). What we did not find in the literature we consulted, and claim only as packaging, is the reading of (2): the cone is cut out by the equilibrium face (through $N_{K_j}(x_j^*)$ and $F_j(x^*)$, both determined by (F, K)) and by the marginal externalities $\nabla_{x_j} u_i(x^*)$, which are *not* determined by (F, K) (Proposition 5.3). The interior case of (d) is also the mechanism behind the theorem of Dubey [8] that Nash equilibria of smooth games are, in general position, Pareto-inefficient: at an interior x^* , any $\lambda \in \Lambda_1(x^*) \setminus \{0\}$ must annihilate every weighted externality, which a generic game does not allow; Example 5.5(i) is an instance.

Proposition 5.3 ((F, K) does not determine welfare). *Let the u_i be continuous on $K = \prod K_i$ with K_i compact, and let x^* be a Nash equilibrium. Define $v_i(x_{-i}) = \max_{y_i \in K_i} u_i(y_i, x_{-i})$ and $u'_i = u_i - v_i$. Then the game (u'_i) has the same best-response correspondences and the same Nash equilibria as (u_i) ; wherever $\nabla_{x_i} u_i$ exists, $\nabla_{x_i} u'_i = \nabla_{x_i} u_i$, so the two games have the same equilibrium operator; and x^* maximises every $W'_\lambda = \sum_i \lambda_i u'_i$, $\lambda \in \mathbb{R}_+^N$, over K : $\Lambda'(x^*) = \mathbb{R}_+^N$. Consequently welfare optimality of an equilibrium is not a property of (F, K) : every pair (F, K) that arises from a game is compatible with $\Lambda(x^*) = \mathbb{R}_+^N$, and the pair of Example 5.5(i) is also compatible with $\Lambda(x^*) = \{0\}$. [proved here; the function $\sum_i (v_i - u_i)$ is the Nikaidô–Isoda gap function [22]]*

Proof. v_i does not depend on x_i , so subtracting it changes neither $\arg \max_{y_i} u_i(y_i, x_{-i})$ nor $\nabla_{x_i} u_i$. By construction $u'_i \leq 0$ on K , and $u'_i(x^*) = 0$ because x_i^* is a best response to x_{-i}^* ; hence $W'_\lambda \leq 0 = W'_\lambda(x^*)$ for every $\lambda \geq 0$. The last sentence combines this with Example 5.5(i), where $\Lambda(x^*) = \{0\}$. $\square$

The modified game need not be C^1 (v_i is in general only directionally differentiable), so it may fall outside Definition 2.3; $\Lambda'(x^*)$ is then understood in the exact sense of Theorem 5.1(a), which needs no gradients. When the best response $b_i(x_{-i})$ is unique for every x_{-i} (e.g. u_i strictly concave in x_i), Danskin's theorem [7] gives $v_i \in C^1$ with $\nabla v_i(x_{-i}) = \nabla_{x_{-i}} u_i(b_i(x_{-i}), x_{-i})$; the modified game is then in the class of Definition 2.3 and, since $b_i(x_{-i}^*) = x_i^*$, it has *zero marginal externalities at x^** : $\nabla_{x_j} u'_i(x^*) = 0$ for $j \neq i$. By Theorem 5.1(d), $\Lambda'_1(x^*) = \mathbb{R}_+^N$ as well, which is the first-order reason for the proposition.

Example 5.4 (Two players, exact cone). Let $K = [0, 1]^2$, $\alpha > 1$, $\beta_1, \beta_2 \geq 0$, and

$$u_1(x) = \alpha x_1 - \frac{1}{2} x_1^2 - \beta_1 x_2, \quad u_2(x) = \alpha x_2 - \frac{1}{2} x_2^2 - \beta_2 x_1.$$

Each player's action harms the other linearly. The equilibrium operator is $F(x) = (x_1 - \alpha, x_2 - \alpha)$, with Jacobian I , so F is 1-strongly monotone and 1-Lipschitz and does not depend on β ; the unique equilibrium is $x^* = (1, 1)$, at which $N_K(x^*) = \mathbb{R}_+^2$ and $F(x^*) = (1 - \alpha)\mathbf{1}$. Here $G(x^*) = \begin{pmatrix} \alpha-1 & -\beta_1 \\ -\beta_2 & \alpha-1 \end{pmatrix}$, so by (2)

$$\Lambda_1(x^*) = \{\lambda \geq 0 : (\alpha - 1)\lambda_1 \geq \beta_2 \lambda_2, (\alpha - 1)\lambda_2 \geq \beta_1 \lambda_1\}.$$

Both payoffs are jointly concave (Hessians $-\text{diag}(1, 0)$ and $-\text{diag}(0, 1)$), so $\Lambda(x^*) = \Lambda_1(x^*)$ by Theorem 5.1(c). Direct check: W_λ is separable, $x_1 = 1$ maximises $\lambda_1(\alpha x_1 - \frac{1}{2} x_1^2) - \lambda_2 \beta_2 x_1$ on $[0, 1]$ iff its derivative at 1, $\lambda_1(\alpha - 1) - \lambda_2 \beta_2$, is ≥ 0 , and symmetrically. Thus the cone is a wedge, non-trivial iff $\beta_1 \beta_2 \leq (\alpha - 1)^2$ and two-dimensional iff the inequality is strict; for $\alpha = 2$, $\beta_1 = \beta_2 = \frac{1}{2}$ it is $\{\frac{1}{2} \lambda_2 \leq \lambda_1 \leq 2 \lambda_2\}$ (Figure 1): the corner equilibrium is welfare-optimal exactly when neither player is weighted more than twice the other. Two games with β and β' have the same F and different cones, which is Proposition 5.3 in miniature; here the modification of that proposition is explicit and C^1 : $v_1(x_2) = \alpha - \frac{1}{2} - \beta_1 x_2$, so $u'_1 = \alpha x_1 - \frac{1}{2} x_1^2 - \alpha + \frac{1}{2}$ no longer depends on β and the modified cone is $\mathbb{R}_+^2$.

Example 5.5 (Cournot duopoly). Let $u_i(q) = q_i(a - q_1 - q_2)$ on $K = [0, 1]^2$; $\partial^2 u_1 / \partial q_1^2 = -2 < 0$, so Lemma 2.4 applies, and $F(q) = (2q_1 + q_2 - a, q_1 + 2q_2 - a)$ has symmetric part $\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \succ 0$. (i) For $a = 1$ the equilibrium $q^* = (\frac{1}{3}, \frac{1}{3})$ is interior, $F(q^*) = 0$, and $\partial u_1 / \partial q_2 = -q_1^* \neq 0$: by Theorem 5.1(d) the interior condition fails for every $\lambda \neq 0$, so $\Lambda(q^*) = \{0\}$ (Dubey's inefficiency, Remark 5.2). Indeed the collusive $(\frac{1}{4}, \frac{1}{4})$ gives each firm $\frac{1}{8} > \frac{1}{9}$. This is an exact potential game with $\Pi(q) = a(q_1 + q_2) - q_1^2 - q_2^2 - q_1 q_2$, whose maximiser is the equilibrium: the potential is not a welfare functional. (ii) For $a = 4$ the equilibrium is the corner $(1, 1)$, $F(q^*) = (-1, -1)$, $G(q^*) = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$ and $\Lambda_1(q^*) = \{\lambda_1 = \lambda_2 \geq 0\}$, a ray. The payoffs are not jointly concave, so Theorem 5.1(c) is silent; but $W_{(1,1)}(q) = (q_1 + q_2)(4 - q_1 - q_2)$ is concave, so $\Lambda(q^*) = \Lambda_1(q^*)$ nevertheless. Concavity is sufficient, not necessary.

Example 5.6 (Three players: a non-polyhedral cone). Let $K = [0, 1]^3$ and

$$u_1 = x_1, \quad u_2 = x_2, \quad u_3 = x_3 - \frac{1}{2} x_1^2 + \frac{1}{4} x_1 + \frac{1}{2} x_1 x_2 - \frac{1}{2} x_2.$$

Each payoff is linear, hence concave, in the player's own variable, so the game is in the class of Definition 2.3; $F \equiv (-1, -1, -1)$, $x^* = (1, 1, 1)$ is the unique Nash equilibrium and $N_K(x^*) = \mathbb{R}_+^3$. Since $\nabla u_3(x^*) = (-\frac{1}{4}, 0, 1)$, $G(x^*)^\top \lambda = (\lambda_1 - \frac{1}{4} \lambda_3, \lambda_2, \lambda_3)$ and $\Lambda_1(x^*) = \{\lambda \geq 0 : \lambda_1 \geq \frac{1}{4} \lambda_3\}$. We compute $\Lambda(x^*)$ exactly. On $\{\lambda_3 = 0\}$, $W_\lambda = \lambda_1 x_1 + \lambda_2 x_2$ is maximised at x^* , so $\mathbb{R}_+^2 \times \{0\} \subseteq \Lambda(x^*)$; as $\Lambda(x^*)$ is a cone it suffices to fix $\lambda_3 = 1$. Then W_λ is increasing in x_3 , so $x_3 = 1$, and affine in x_2 ,

so the maximum over x_2 is attained at $x_2 \in \{0, 1\}$. For $x_2 = 1$, $W_\lambda = \frac{1}{2} + \lambda_2 + (\lambda_1 + \frac{3}{4})x_1 - \frac{1}{2}x_1^2$ is maximised on $[0, 1]$ at $x_1 = 1$ iff $\lambda_1 \geq \frac{1}{4}$, with value $W_\lambda(x^*) = \lambda_1 + \lambda_2 + \frac{3}{4}$. For $x_2 = 0$, $W_\lambda = 1 + (\lambda_1 + \frac{1}{4})x_1 - \frac{1}{2}x_1^2$ has maximum $1 + \frac{1}{2}(\lambda_1 + \frac{1}{4})^2$ if $\lambda_1 \leq \frac{3}{4}$ (at $x_1 = \lambda_1 + \frac{1}{4}$) and $\lambda_1 + \frac{3}{4} \leq W_\lambda(x^*)$ otherwise. Hence $\lambda \in \Lambda(x^*)$ iff $\lambda_1 \geq \frac{1}{4}$ and $\lambda_1 + \lambda_2 + \frac{3}{4} \geq 1 + \frac{1}{2}(\lambda_1 + \frac{1}{4})^2$, i.e.

$$\Lambda(x^*) \cap \{\lambda_3 = 1\} = \{\lambda_1 \geq \frac{1}{4}, \lambda_2 \geq \frac{1}{2}(\frac{3}{4} - \lambda_1)^2\}.$$

Its boundary contains the parabolic arc $\lambda_2 = \frac{1}{2}(\frac{3}{4} - \lambda_1)^2$, $\frac{1}{4} \leq \lambda_1 \leq \frac{3}{4}$, which no finite union of segments contains (a line meets a parabola in at most two points); since the slice of a polyhedral cone by a hyperplane is a polyhedron, $\Lambda(x^*)$ is *not polyhedral*, and $\Lambda(x^*) \subsetneq \Lambda_1(x^*)$ with a curved boundary. For instance $\lambda = (0.3, 0, 1) \in \Lambda_1 \setminus \Lambda$: the point $(0.55, 0, 1)$ improves W_λ by 0.10125. The degeneracy $F \equiv \text{const}$ is inessential: subtracting $\frac{\varepsilon}{2}x_i^2$ from each own payoff makes F ε -strongly monotone, keeps x^* for $\varepsilon < 1$, and the boundary of the slice remains curved (E2c in Section 6, numerically). [proved here; the counterexample was pointed out in the internal review of v0.1]

Example 5.7 (Two commodities: $\Lambda \neq \Lambda_1$ in routing). Theorem 5.1(a),(b) use neither the Nash property nor the product structure: they hold for any finite family of C^1 functions u_k on K and any $x^* \in K$, with Λ and Λ_1 defined by the same formulas. Apply them to the costs $C_k(f) = \sum_{p \in P_k} f_p c_p(f)$ borne by the commodities of a network, $u_k = -C_k$, at a Wardrop equilibrium f^* . Take two commodities with unit demands: commodity 1 chooses between a private edge of cost $f + 2$ and a shared edge of cost f ; commodity 2 between a private edge of cost $f/4$ and the same shared edge. With $y = (f_B, f_D) \in [0, 1]^2$ the flows sent to the shared edge (path flows are determined by edge flows here, so f^* is a well-defined point),

$$C_1(y) = (1 - f_B)(3 - f_B) + f_B(f_B + f_D), \quad C_2(y) = \frac{1}{4}(1 - f_D)^2 + f_D(f_B + f_D),$$

both with indefinite Hessians (determinant -1 , from the shared-edge cross term), so (c) is unavailable. The Wardrop equilibrium is $f^* = (1, 0)$ (shared path cost $1 \leq 2$ for commodity 1, private cost $\frac{1}{4} \leq 1$ for commodity 2), unique because the reduced operator $y \mapsto (c_B - c_A, c_D - c_C)$ has Jacobian $\begin{pmatrix} 2 & 1 \\ 1 & 5/4 \end{pmatrix} \succ 0$; $C(f^*) = (1, \frac{1}{4})$. At f^* , $N_K(f^*) = \mathbb{R}_+ \times \mathbb{R}_-$, $\nabla C_1(f^*) = (0, 1)$ and $\nabla C_2(f^*) = (0, \frac{1}{2})$, so $-\lambda_1 \nabla C_1 - \lambda_2 \nabla C_2 = (0, -\lambda_1 - \frac{1}{2}\lambda_2) \in N_K(f^*)$ for every $\lambda \geq 0$: $\Lambda_1(f^*) = \mathbb{R}_+^2$. For Λ , fix $\lambda_2 = 1$ ($\lambda_2 = 0$ gives $\lambda \in \Lambda$: C_1 is minimised at f^*). The Hessian of $\lambda_1 C_1 + C_2$ is $\begin{pmatrix} 4\lambda_1 & \lambda_1 + 1 \\ \lambda_1 + 1 & 5/2 \end{pmatrix}$, positive semidefinite iff $10\lambda_1 \geq (\lambda_1 + 1)^2$, in which case the first-order condition suffices and $\lambda \in \Lambda$. Otherwise the minimum over the box lies on an edge, where the restriction is a convex quadratic: on $f_B = 1$ and on $f_D = 0$ it is attained at f^* (derivatives $\lambda_1 + \frac{1}{2} + \frac{5}{2}t > 0$ and $4\lambda_1(f_B - 1) \leq 0$); on $f_D = 1$ the value is at least $\frac{15}{8}\lambda_1 + 1 > \lambda_1 + \frac{1}{4}$; on $f_B = 0$ it is $3\lambda_1 + \frac{1}{4}(1 - t)^2 + t^2$, minimised at $t = \frac{1}{5}$ with value $3\lambda_1 + \frac{1}{5}$. Hence f^* is optimal iff $\lambda_1 + \frac{1}{4} \leq 3\lambda_1 + \frac{1}{5}$, and

$$\Lambda(f^*) = \{\lambda \geq 0 : \lambda_2 \leq 40\lambda_1\} \subsetneq \Lambda_1(f^*) = \mathbb{R}_+^2.$$

For $\lambda = (0, 1)$, the flow $y = (0, \frac{1}{5})$ reroutes commodity 1 to its private edge and sends a fifth of commodity 2 through the shared edge, lowering C_2 from $\frac{1}{4}$ to $\frac{1}{5}$ (while C_1 rises from 1 to 3). The first-order cone misses this because f^* is a corner at which the KKT conditions hold for every weighting, while the per-commodity costs are non-convex in path flows. (f^* is also the utilitarian optimum.) [proved here; verified in E5]

6 Numerical checks

All computations use the projected-gradient iteration of Theorem 3.5(b) with $\gamma = \mu/L^2$, run to a fixed-point step below 10^{-12} , and report the natural residual $\|x - \Pi_K(x - F(x))\|$, which vanishes exactly at solutions (Lemma 3.4). The protocols and the pass/fail criteria of each experiment are stated in the script docstrings, reproduced in Appendix B; each script evaluates

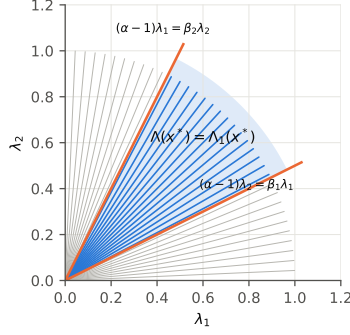


Figure 1: Example 5.4, $\alpha = 2$, $\beta_1 = \beta_2 = 0.5$: shaded, the closed-form wedge; orange, its boundaries (26.57° , 63.43°); every fifth of the 181 tested directions, blue if the exact maximisation of W_λ returns x^* , grey otherwise (agreement 181/181).

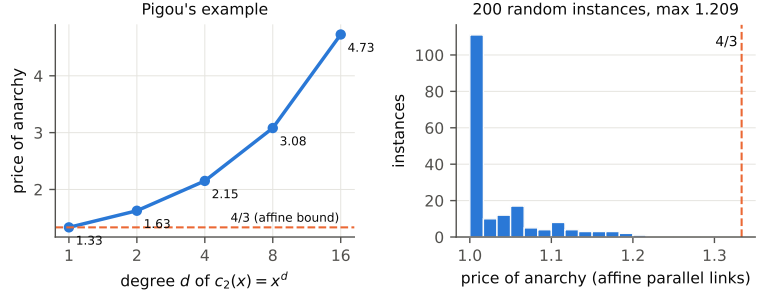


Figure 2: E1. Left: price of anarchy of Pigou’s example against d (values in `results/tables_traffic.md`); dashed, the affine bound $4/3$ of Theorem 3.8. Right: price of anarchy of 200 random affine parallel-link instances; none exceeds $4/3$.

its own criteria and writes the flags and margins into its JSON output, and the phrases “all criteria hold” below are generated from those flags. There is no external pre-registration record: the criteria were written in the same session as the code and before the reference run, and we can offer no stronger evidence than that. The experiments verify theorems and produce descriptive statistics of one sampling distribution; they discover nothing that is not proved above.

E1: routing (Pigou, affine price of anarchy, tolls). all 5 criteria hold. Pigou: PoA = 1.333333 for $d = 1$, 1.6258 for $d = 2$ and 4.7268 for $d = 16$; closed form and grid search agree to 1.5×10^{-11} (Figure 2). Affine parallel-link instances: maximum PoA 1.2094, 0 violations of the bound $4/3$ (Figure 2); 86 of the 200 instances (43%) have PoA = 1 exactly (equilibrium and optimum coincide, e.g. when every $b_e = 0$), which pulls the mean (1.0305) and the median (1.0027) towards 1; 41% of the instances have PoA > 1.01 and 12% have PoA > 1.10; maximum residual 3.4×10^{-11} , maximum deviation of the solver from exact water-filling 8.8×10^{-11} , at most 2649 iterations. E1c: the equilibrium of the tolled operator $c + \tau$ computed by the solver coincides with the optimum computed independently by water-filling on the marginal costs to 8.8×10^{-11} in every instance. (For affine costs $c + \tau$ and ∇C are the same function, so this checks the solver against an independent computation, not the identity $c + \tau = \nabla C$.)

E2–E4: welfare-weight cones. all 12 criteria hold. (E2) Example 5.4: $x^* = (1, 1)$ with residual 0; closed form, linear test and exact box maximisation agree on 181/181 directions, 73 of which lie in the cone; for $\lambda = (1, 3)$ (outside the wedge) the exact maximiser is $x = (0.5, 1)$, better by 0.1250. Cournot (i): $q^* = (0.3333, 0.3333)$, 0/61 grid directions in Λ ; (ii): $q^* = (1, 1)$, cone of dimension 1, agreement 61/61. Nikaidô–Isoda modification: maximum of W'_λ over the grid $\leq 2.8 \times 10^{-17}$ for all directions (criterion met: yes). (E2c) Example 5.6: residual 0 at x^* ; on 7381 points of the slice $\lambda_3 = 1$ the exact test and the closed-form parabola disagree on 0 points; the boundary found by bisection is within 1.0×10^{-8} of the parabola; $\lambda = (0.3, 0, 1)$ is in Λ_1 , not in Λ , with witness $(0.55, 0, 1)$ and gap 0.10125. In the strongly monotone variant ($\varepsilon = 0.2$, $\mu = 0.2$) the bisected boundary at 9 values $\lambda_1 \in [0.35, 0.75]$ has second differences that are all positive (minimum absolute value 1.7×10^{-3}), so it is not piecewise linear on that range. (E3–E4, Table 1) In the 200 concave games, 61 (30%) have a non-trivial cone; all 268/268 cone points and all 882/882 in-cone grid points are in Λ , and all 24718/24718 out-of-cone grid

N	conc.	games	$\Lambda_1 \neq \{0\}$	$\dim \Lambda_1$: games	cone pts $\in \Lambda$	grid $\in \Lambda_1$: $\in \Lambda$	grid $\notin \Lambda_1$: $\notin \Lambda$	witn.
3	yes	100	28	1:4, 2:5, 3:19	115/115	575/575	8525/8525	0
4	yes	100	33	1:2, 2:7, 3:15, 4:9	153/153	307/307	16193/16193	0
3	no	100	57	1:13, 2:14, 3:30	162/206	1025/1063	8037/8037	23
4	no	100	46	1:2, 2:2, 3:13, 4:29	152/246	1265/1337	15163/15163	35

Table 1: E3–E4. Random monotone quadratic games on $[0, 1]^N$. “conc.” is joint concavity; “cone pts” are vertices and barycentre of $\Lambda_1 \cap \{\mathbf{1}^\top \lambda = 1\}$; “witn.” counts games with a certified $\lambda \in \Lambda_1 \setminus \Lambda$ among the cone points (a lower bound). Concave games realise $\Lambda = \Lambda_1$ in every test; non-concave games realise the strict inclusion in a sizeable fraction.

points are outside Λ : equality $\Lambda = \Lambda_1$ in every tested direction, as Theorem 5.1(c) predicts (for a trivial cone only the exclusion is tested). In the 200 non-concave games, 103 have a non-trivial cone; the inclusion $\Lambda \subseteq \Lambda_1$ is never violated (23200/23200), but only 314/452 cone points and 2290/2400 in-cone grid points belong to Λ , and 58 instances (56% of those with a non-trivial cone) carry a certified witness of strict inclusion (for instance, in game N3 nonconcave with $x^* = (1.000, 1.000, 1.000)$ and $\lambda = (0.000, 0.552, 0.448) \in \Lambda_1$, the point $(0.000, 1.000, 1.000)$ improves W_λ by 0.215). Witnesses are sought only among the cone points, so 56% is a *lower bound* on the fraction of instances with $\Lambda \subsetneq \Lambda_1$: the 110 in-cone grid points that fail the exact test are not counted. Maximum residuals 7.7×10^{-12} (concave) and 2.1×10^{-12} (non-concave); 51920 exact box maximisations in total.

E5: commodity weights in two-commodity routing. all 5 criteria hold. Example 5.7: the solver returns $f^* = (1, 0)$ with residual 0 and costs $(1, 0.25)$; all 91 tested directions lie in Λ_1 and 89 in Λ ; bisection on $\lambda_2 = 1$ puts the boundary of Λ at $\lambda_1 = 0.025000$ ($1/40.000$, the predicted $1/40$); for $\lambda = (0, 1)$ the exact minimiser is $(0, 0.2)$ with costs $(3, 0.2)$, a gap of 0.0500. In 200 random two-commodity instances (same seed), 33 have corner equilibria, the direction grid meets Λ_1 in 90 and Λ_1 is the whole orthant in 9; 21 instances (23% of those where the grid meets Λ_1) carry a certified witness $\lambda \in \Lambda_1 \setminus \Lambda$, and no direction certified in Λ lies outside Λ_1 (0 violations; maximum residual 3.8×10^{-12}).

7 What can be claimed

Table 2 lists every claim made in this note with its status. The reader in a hurry should retain three lines: the classical results are theorems and are proved above; the “formal relationship” between equilibrium and welfare is the first-order coincidence of Proposition 4.1, which is a definition rather than a mechanism; and (F, K) alone carries no welfare information (Proposition 5.3), so any welfare statement needs the externalities as separate data.

8 Limitations and open questions

Limitations. (1) Everything is finite-dimensional, deterministic and C^1 ; mixed strategies, atomic congestion games, dynamics and stochastic demand are outside the note. (2) Theorem 5.1 is a first-order statement; its only non-trivial sufficient condition for equality is concavity, which excludes Cournot-type payoffs and multi-commodity routing costs. (3) The numerical experiments verify theorems on quadratic games over boxes and on small networks; the reported frequencies describe one sampling distribution and carry no general meaning. (4) Deciding $\lambda \in \Lambda(x^*)$ for a non-concave quadratic W_λ on a box is a non-convex quadratic programme, tractable here only because $N \leq 4$. (5) We have not surveyed the toll-pricing and multi-objective literature exhaustively (Remark 5.2).

Claim	Status
Symmetric Jacobian on a convex open set gives a potential (Prop. 3.1).	classical; proved here
VI solutions of a gradient field are the stationary points of the potential, its minimisers if convex (Thm. 3.2).	classical (BMW 1956, Monderer–Shapley 1996); proved here
Nash and Wardrop equilibria are VI solutions (Lemmas 2.4, 2.7).	classical; proved here
Strict monotonicity: uniqueness; strong monotonicity and Lipschitz: existence and linear convergence (Thm. 3.5).	classical (Kinderlehrer–Stampacchia, Facchinei–Pang); proved here
Pigou: PoA = 4/3 for $d = 1$, unbounded in d (Ex. 3.7); PoA $\leq 4/3$ for affine costs (Thm. 3.8).	classical (Roughgarden–Tardos 2002); proved here; checked on 200 instances (E1)
Equilibrium is a W -maximiser only if $\nabla W(x^*), -F(x^*) \in N_K(x^*)$; iff when W is concave (Prop. 4.1).	proved here; elementary
Marginal-cost tolls make the cost minimisers the equilibria; toll set = $-F(\bar{x}) - N_K(\bar{x})$ (Cor. 4.2).	classical (Pigou, Knight, BMW, Hearn–Ramana; inverse optimisation); proved here; solver checked against water-filling (E1c)
$\Lambda(x^*)$ is a closed convex cone, $\Lambda \subseteq \Lambda_1$, equality under joint concavity, block form, Pareto link (Thm. 5.1).	proved here; (a),(e) standard scalarisation (weight set)
(F, K) alone does not determine $\Lambda(x^*)$ (Prop. 5.3).	proved here; repackaging of Nikaidô–Isoda; Danskin for the C^1 case
Exact wedge cone in the two-player example (Ex. 5.4).	proved here; verified (E2)
$\Lambda(x^*)$ is not polyhedral in general (Ex. 5.6).	proved here (pointed out in internal review); verified (E2c)
$\Lambda \neq \Lambda_1$ for two commodities with affine costs (Ex. 5.7).	proved here; verified (E5)
$\Lambda = \Lambda_1$ in random concave games.	theorem; verified in every tested direction of 200 games (E3)
$\Lambda \subsetneq \Lambda_1$ occurs without joint concavity.	certified witnesses in 58 games (E4), at least 56% of those with a non-trivial cone; frequency descriptive only
A new equilibrium concept or a general equilibrium solution.	not claimed
Open questions 8.1, 8.2.	open (reformulated after the review)

Table 2: Claims and their status.

Open question 8.1 (Structure of $\Lambda(x^*)$ beyond first order). For polynomial payoffs on a polyhedral K , $\Lambda(x^*) = \{\lambda \geq 0 : \forall x \in K, W_\lambda(x) \leq W_\lambda(x^*)\}$ is defined by a first-order formula over the reals, hence is semialgebraic by the Tarski–Seidenberg theorem [4]: a finite description by polynomial inequalities always exists (for quadratic payoffs on a box, $\Lambda = \bigcap_\Phi \{\lambda : W_\lambda(x^*) \geq \max_\Phi W_\lambda\}$ over the 3^N faces Φ , the parabola of Example 5.6 being the term of the face $\{x_2 = 0, x_3 = 1\}$), and for $N = 2$ every closed convex cone of $\mathbb{R}^2$ is polyhedral. Example 5.6 shows that for $N = 3$ the cone need not be polyhedral. What remains: (i) characterise, in terms of the active-face pattern of x^* and the Hessians, when $\Lambda(x^*) = \Lambda_1(x^*)$ and when $\Lambda(x^*)$ is polyhedral without joint concavity; (ii) determine the complexity of deciding $\lambda \in \Lambda(x^*)$ for quadratic payoffs on a box (it contains the certification that x^* is a global maximiser of an indefinite quadratic over the box; already deciding local optimality of a point in an indefinite quadratic programme is NP-hard [19]), and whether a description of size polynomial in N exists when F is strongly monotone.

Open question 8.2 (Commodity weights in routing). In a multi-commodity network let $C_k(f) = \sum_{p \in P_k} f_p c_p(f)$ and let f^* be a Wardrop equilibrium *in path flows*; edge flows are unique for strictly increasing costs but path flows are not, and $\Lambda(f^*)$ depends on the representative chosen. Theorem 5.1(a),(b) give $\Lambda(f^*) \subseteq \Lambda_1(f^*)$, with equality when no edge is shared between commodities (each C_k is then convex for affine costs), and Example 5.7 shows that $\Lambda \neq \Lambda_1$ already for two commodities, one shared edge and affine costs. Characterise $\Lambda(f^*)$ for $m \geq 3$ commodities and affine costs (by analogy with Example 5.6 we expect it to be non-polyhedral in

general) and identify the network structures for which $\Lambda = \Lambda_1$.

Next steps. Attempt the case analysis of Question 8.1(i) for $N = 3$ on the 27 faces of the box, with Example 5.6 as the guiding case; run E4 and E5 with a global solver for larger N and m ; and confirm with the author whether the “formulation reviewed” in the line’s record was an instance of Proposition 4.1 or of Corollary 4.2, which is our reading.

A Reproducibility

The directory contains `experiments/vi_solver.py` (projections and the iteration of Theorem 3.5(b)), `experiments/traffic_poa.py` (E1), `experiments/welfare_cone.py` (E2–E4, including E2c) and `experiments/commodity_cone.py` (E5). Seed 20260930, consumed sequentially by a single generator per script (reproducible, but reordering the experiments changes the instances); Python 3.11.15, NumPy 2.4.6 [13], SciPy 1.17.1 [28] (`linprog` with the HiGHS method for the cone geometry); E1 took 2 s, E2–E4 86 s and E5 4 s on shared cores. The scripts write `results/results_*.json` (including a `criteria` block with pass/fail flags) and Markdown tables; `experiments/make_numbers.py` converts them into the macros and table bodies used here, so that every number in the text is generated.

Solver. Projection onto the box is coordinatewise clipping; projection onto the simplex uses the sort-based exact algorithm. The step is $\gamma = \mu/L^2$ with μ the least eigenvalue of the symmetric part of the Jacobian and L its spectral norm ($\mu = \min a_e$, $L = \max a_e$ for parallel links); the iteration stops when the step is below 10^{-12} (E1) or 10^{-13} (E2–E5). In Example 5.6 F is constant ($\mu = L = 0$) and the fixed-point map with $\gamma = 1$ is used; the residual at x^* certifies the equilibrium.

Face enumeration. For each of the 3^N patterns in $\{0, 1, \text{free}\}^N$ the free coordinates solve $H_{ff}x_f = h_f - H_{fb}x_b$; candidates outside $[-10^{-9}, 1 + 10^{-9}]$ are discarded; the maximum over candidates is the maximum over the box, because a maximiser lies in the relative interior of some face and is stationary for the restriction. If H_{ff} is singular the system is either inconsistent (no stationary point in the relative interior of that face, so the maximum over the face is attained on a lower-dimensional face, also enumerated) or consistent with W_λ constant along the null directions (the same value is attained on a lower-dimensional face); the least-squares candidate used in that case is therefore redundant and the enumeration remains exhaustive.

Boundary detection. A coordinate within 10^{-9} of 0 or 1 is active (the projection returns exact bounds).

B Protocols and criteria

E1. E1a: Example 3.7 in closed form and by a grid search over the split (2×10^5 points). E1b: 200 parallel-link networks with 2–5 links, $a_e \sim U(0.05, 1)$, $b_e = 0$ with probability $\frac{1}{2}$ and otherwise $U(0, 1)$, unit demand; equilibrium ($F = c$) and optimum ($F = \nabla C = 2a \circ f + b$) by the solver on the simplex and, independently, by exact water-filling on the common (marginal) cost level. E1c: the tolled operator $c + \tau$, $\tau_e = a_e f_e$ (Corollary 4.2), by the solver, against the water-filling optimum. *Criteria:* residuals $\leq 10^{-8}$; solver vs. water-filling $\leq 10^{-7}$; PoA $\leq 4/3 + 10^{-9}$ in every instance and $4/3$ to 10^{-12} for Pigou with $d = 1$; tolled equilibrium vs. water-filling optimum $\leq 10^{-7}$.

E2–E4. *Games:* N players, scalar actions in $[0, 1]$, $u_i(x) = -\frac{1}{2}x^\top H_i x + h_i^\top x$, so $F(x) = Mx - m$ with $M_{ij} = (H_i)_{ij}$; concave family $H_i = B_i B_i^\top$ with Gaussian B_i , non-concave family H_i Gaussian symmetric; in both, the diagonals are raised by a common amount so that the symmetric part of M has least eigenvalue $\frac{1}{2}$ (strong monotonicity, unique equilibrium, own-concavity); $h_i \sim N(1, 1.5^2)$, which puts 64–75% of the equilibrium coordinates on the boundary. *Tests:* $\lambda \in \Lambda_1$ is the linear system of Theorem 5.1(d) with $N_{[0,1]}(0) = \mathbb{R}_-$, $N_{[0,1]}(1) = \mathbb{R}_+$ and $\{0\}$ inside, checked to 10^{-9} ; $\lambda \in \Lambda$ iff $W_\lambda(x^*) \geq \max_K W_\lambda - 10^{-8}$, with the maximum by face enumeration (Appendix A), cross-checked against multistart projected gradient ascent on 13 concave instances (discrepancy 1.4×10^{-17}); a point with $W_\lambda(x) > W_\lambda(x^*) + 10^{-6}$ is a *certified witness*. *Geometry of Λ_1 :* non-triviality and dimension by linear programming (implicit equalities row by row); up to 12 vertices of $\Lambda_1 \cap \{\mathbf{1}^\top \lambda = 1\}$ plus their barycentre are the *cone points*; every point of a simplex grid (step $1/12$ for $N = 3$, $1/8$ for $N = 4$) is classified by both tests. *Criteria:* (E2) closed form = linear = exact on all 181 directions of Example 5.4; no grid direction in Λ for Example 5.5(i); linear = exact on all grid directions for (ii); $\max W'_\lambda \leq 10^{-12}$ on a 401×401 grid of K and 61 directions for Proposition 5.3. (E2c) exact test = closed form of Example 5.6 on a 121×61 grid of the slice $\lambda_3 = 1$; bisected boundary within 10^{-7} of the parabola; $(0.3, 0, 1) \in \Lambda_1 \setminus \Lambda$ with gap 0.10125 to 10^{-9} . (E3) every cone point and every grid point in Λ_1 is in Λ , every grid point outside Λ_1 is outside Λ . (E4) the latter still holds (Theorem 5.1(b)); witnesses are counted with no prediction. All residuals $\leq 10^{-8}$.

E5. The network of Example 5.7 and 200 random instances of the same shape ($a \sim U(0.05, 1)$; $b = 0$ with probability $\frac{1}{2}$, else $U(0, 1)$; unit demands), solved by the projected-gradient iteration for the reduced operator on $[0, 1]^2$; Λ_1 by the linear test and Λ by face enumeration of $\lambda_1 C_1 + \lambda_2 C_2$ on 91 directions $\lambda = (\cos \theta, \sin \theta)$. *Criteria:* residuals $\leq 10^{-8}$; fixed instance: all directions in Λ_1 , boundary ratio $1/40$ to 10^{-6} , witness gap $1/20$ to 10^{-9} for $\lambda = (0, 1)$; random instances: no direction certified in Λ (gap $\leq 10^{-11}$) outside Λ_1 . Directions accepted by the exact test only within its 10^{-8} tolerance while failing the linear test are recorded as tolerance-level disagreements (3 in the reference run, largest first-order violation 5.1×10^{-5}).

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